

# Pemphigoid group

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# Pemphigoid group

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autoimmune sub-epidermal/ sub-epithelial blistering disease.

members of the 'Pemphigoid ' family (6 )

1. Bullous pemphigoid (BP)

Variant Pemphigoid gestationis (PG)

2. Cicatricial pemphigoid (CP)

3. Epidermolysis bullosa acquisita (EBA)

4. Linear IgA disease (LAD)

5. Dermatitis herpetiformis (DH)

6. Bullous systemic lupus erythematosus (Bullous SLE)

A



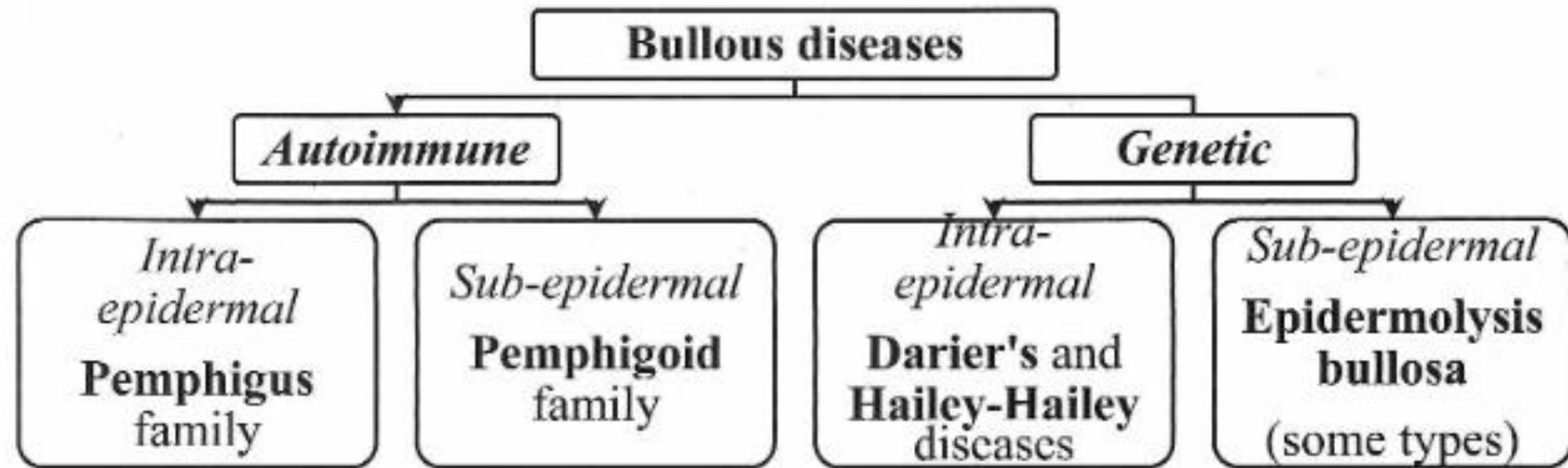
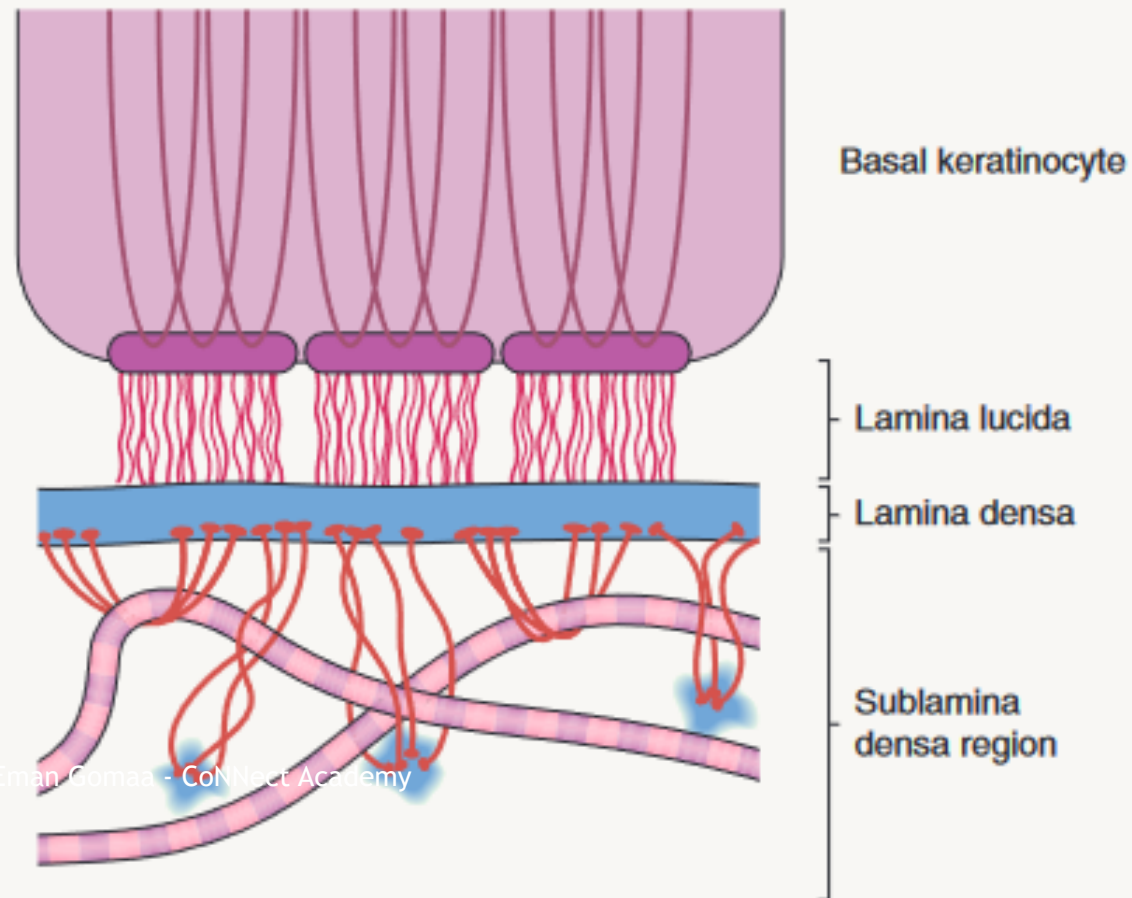


Fig 0.3 Autoimmune and genetic blistering diseases.

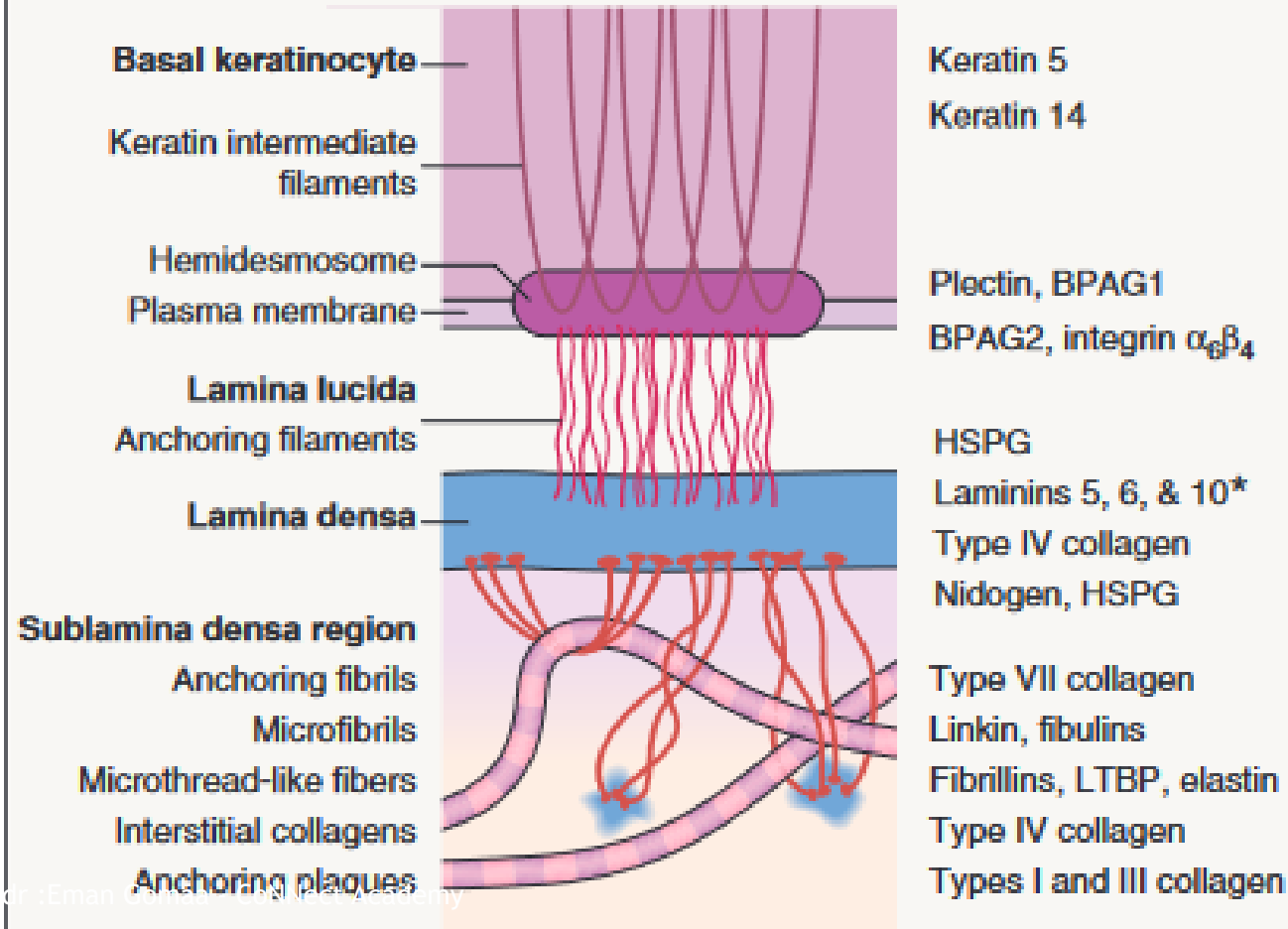
Table 0.1 Autoimmune and genetic blistering diseases.

|                                |            | Level of blister                          |                                    |
|--------------------------------|------------|-------------------------------------------|------------------------------------|
|                                |            | Intra-epidermal                           | Sub-epidermal                      |
| Mechanism of blister formation | Autoimmune | Pemphigus family                          | Pemphigoid family                  |
|                                | Genetic    | Darier's disease<br>Hailey-Hailey disease | Epidermolysis bullosa (some types) |

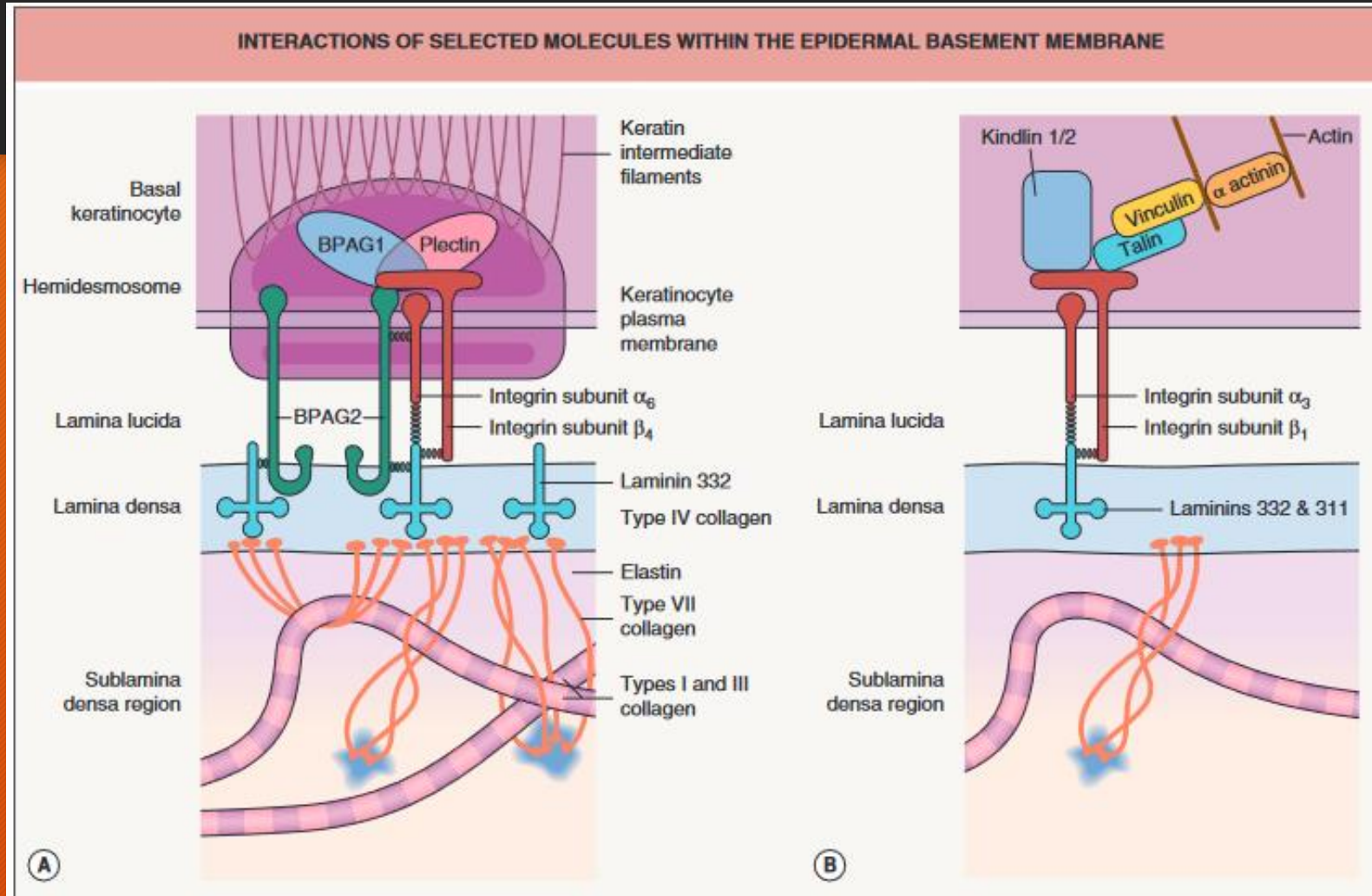
#### FOUR MAJOR ULTRASTRUCTURAL SUBREGIONS OF THE EPIDERMAL BASEMENT MEMBRANE



# “LAMINATED” MODEL OF THE EPIDERMAL BASEMENT MEMBRANE

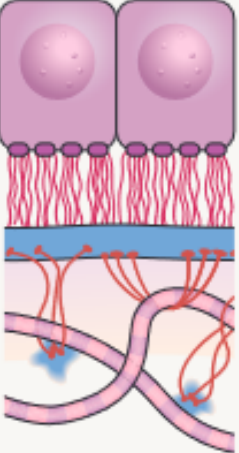






### TARGETS COMMON TO AUTOIMMUNE AND INHERITED BLISTERING DISEASES

| Protein target             | Structural target                    | Autoimmune disease                         | Genetic disease                        |
|----------------------------|--------------------------------------|--------------------------------------------|----------------------------------------|
| BPAG1e                     | HD                                   | BP                                         | Recessive EB simplex                   |
| Type XVII collagen (BPAG2) | HD-anchoring filament complexes      | BP, PG, MMP, linear IgA bullous dermatosis | Junctional EB (often milder)           |
| Integrin subunit $\beta_4$ | HD-anchoring filament complexes      | Ocular MMP                                 | Junctional EB with pyloric atresia     |
| Laminin 332                | Lamina lucida-lamina densa interface | Anti-epiligrin MMP                         | Junctional EB (often more severe)      |
| Type VII collagen          | Anchoring fibrils                    | EB acquisita<br>Bullous eruption of SLE    | Dystrophic EB (dominant and recessive) |

| LOCALIZATION OF AUTOANTIGENS IN SUBEPIDERMAL IMMUNOBULLOUS DISEASES AND MUTATED PROTEINS IN EPIDERMOLYSIS BULLOSA                                                                                                                                                                                                           |                                                                                                                                                                                                                                     |                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Epidermal basement membrane                                                                                                                                                                                                                                                                                                 | Autoantigens in immunobullous diseases                                                                                                                                                                                              | Mutated proteins in epidermolysis bullosa (EB)                                                                                                                                                                              |
|  <p>Basal keratinocytes</p> <p>Keratins 5 &amp; 14</p> <p>Lamina lucida</p> <p>BPAG1, plectin<br/>BPAG2, integrin <math>\alpha_6\beta_4</math></p> <p>Lamina densa</p> <p>Laminin 332</p> <p>Sublamina densa</p> <p>Type VII collagen</p> | <p>Bullous pemphigoid, pemphigoid gestationis<br/>LABD, mucous membrane pemphigoid</p> <p>Anti-epiligrin mucous membrane pemphigoid</p> <p>Epidermolysis bullosa acquisita<br/>Bullous eruption of systemic lupus erythematosus</p> | <p>EB simplex – classic &amp; AR-K14</p> <p>EB simplex with MD, with PA, Ogna &amp; AR-BP230</p> <p>Junctional EB – localized/other &amp; with PA</p> <p>Junctional EB – generalized &amp; inversa</p> <p>Dystrophic EB</p> |



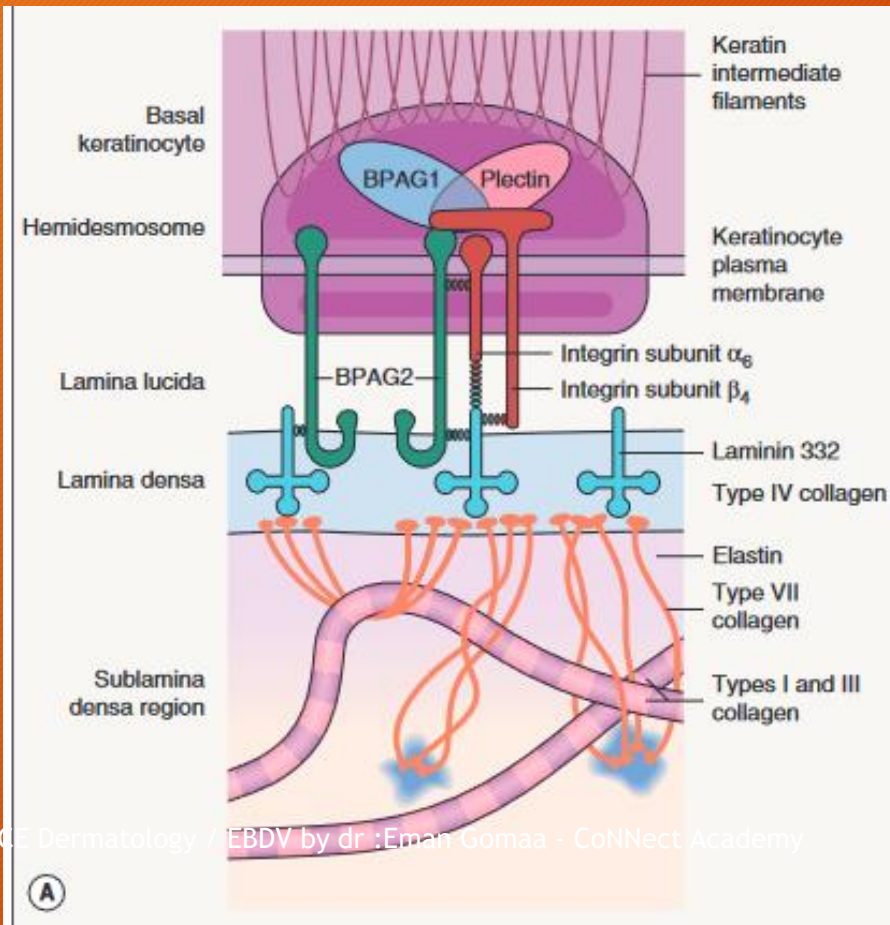
## WELL-CHARACTERIZED SUBEPIDERMAL IMMUNOBULLOUS DISEASES

| Disease                                          | Antigen           | Size (kDa) | Localization                               |
|--------------------------------------------------|-------------------|------------|--------------------------------------------|
| Bullous pemphigoid                               | BPAG1             | 230        | Intracytoplasmic plaque of hemidesmosomes  |
|                                                  | BPAG2             | 180        | Hemidesmosome-anchoring filament complexes |
| Pemphigoid gestationis                           | BPAG2             | 180        | Hemidesmosome-anchoring filament complexes |
| Linear IgA bullous dermatosis                    | BPAG2             | 120→97     | Hemidesmosome-anchoring filament complexes |
| Mucous membrane (cicatricial) pemphigoid         | BPAG2             | 180        | Hemidesmosome-anchoring filament complexes |
| Anti-epiligrin mucous membrane pemphigoid        | Laminin 332       | 400–440    | Lamina lucida–lamina densa interface       |
| EB acquisita                                     | Type VII collagen | 290        | Anchoring fibrils                          |
| Bullous eruption of systemic lupus erythematosus | Type VII collagen | 290        | Anchoring fibrils                          |

**Table 28.4 Well-characterized subepidermal immunobullous diseases.** BPAG, bullous pemphigoid antigen; EB, epidermolysis bullosa.

# Bullous pemphigoid

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- Most common autoimmune subepidermal blistering disease .
- Elderly men
- Aetiopathogenesis:
  - Autoantibody against
    - BpA2(180): NC16A domain ,1<sup>st</sup> extracellular part
    - BPA1(230)



# Clinical feature

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Fig. 30.3 Bullous pemphigoid – urticarial presentation. Multiple, firm annular, arciform and polycyclic urticarial plaques. Note the absence of bullae.



- 1- skin
  - Non bullous phase : pruritis , urticarial lesion +- excoriated eczematous or papular lesions)
  - Bollous phase :tense bullae on erythamatus or normal skin >>> Rupture >>>>PIH
- Symmetrical distribution
- Flexure , lower trunk
- Mucous membrane :uncommon
  - 10;30%(oral cavity)





**Fig. 30.4** Bullous pemphigoid – eczematous presentation. Large pink eczematous plaques on the trunk and upper



**Fig. 30.5** Bullous pemphigoid – nonspecific lesions due to pruritus. Multiple excoriations and nonspecific lesions of prurigo simplex are present. Bullous pemphigoid is in the differential diagnosis of pruritus in the absence of an obvious dermatosis.



### UNUSUAL CLINICAL VARIANTS OF BULLOUS PEMPHIGOID

Dyshidrosiform pemphigoid – palmoplantar vesicles and bullae (Fig. 30.6A,B)

Pemphigoid vegetans – intertriginous vegetating plaques (Fig. 30.6C)

Pemphigoid nodularis – prurigo nodularis-like lesions (see Fig. 6.6)

Vesicular pemphigoid – dermatitis herpetiformis-like presentation with small grouped vesicles

Large erosive TEN-like lesions (Fig. 30.6D)

Papular pemphigoid

Eczematous pemphigoid (Fig. 30.4)

Erythrodermic pemphigoid

Lichen planus pemphigoides (see Ch. 11)

*Localized lesions:*

- pretibial
- distal end of amputated limb\*
- vulvar
- paralyzed limb
- peristomal
- sites of radiotherapy†
- umbilical
- Brunsting-Perry form‡

### • NED 3V 2 L c (gestational )

1. Gestational
2. Nodular
3. Erythrodermic
4. Dyshidrosiform
5. Vesicular
6. Vulvar
7. Vegetans
8. Lp
9. Localized
10. childhood



**Table 3-8.** Pemphigoid Variants

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pemphigoid vegetans                                                 | Vegetative plaques in <b>intertriginous</b> areas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Infantile/childhood Pemphigoid                                      | Frequently p/w <b>acral bullae</b> → generalizes; ↑ <b>facial/genital</b> involvement; clinically indistinguishable from childhood LABD/CBDC → need DIF/IIF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Pemphigoid nodularis                                                | Clinically resembles prurigo nodularis; typically lacks bullae                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lichen Planus pemphigoides                                          | <b>LP/BP overlap</b> syndrome w/circulating antibodies against BP180; p/w LP-like papules/plaques and tense bullae arising on skin unaffected by LP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Pemphigoid gestationis (gestational pemphigoid, herpes gestationis) | Abrupt onset; <b>any trimester (second and third most common)</b> , immediately <b>post-partum</b> , or a/w trophoblastic tumors ( <b>choriocarcinoma</b> , hydatidiform mole); starts as urticarial/vesicular plaques on trunk, abdomen, <b>umbilicus</b> → rapidly generalizes; <b>75% flare at time of delivery</b> ; <b>anti-HLA antibodies (~100%)</b> ; strongly a/w <b>HLA-DR3 (70%), DR4 (50%)</b> , or both (45%); DIF: <b>linear C3 (100%) &gt; linear IgG (30%)</b> ; IIF: only positive in 30%; <b>ELISA for BP180-NC16A</b> is best serum test; ↑risk of <b>premature delivery</b> and SGA neonates; <b>neonates may develop transient blistering (10%)</b> ; <b>recurs</b> in subsequent pregnancies; a/w <b>Graves' disease</b> and anti-thyroid antibodies; Rx: systemic steroids |
| Localized pemphigoid                                                | Pretibial, peristomal, vulvar, umbilical, distal portion of amputated limb ("stump pemphigoid"), radiotherapy sites, paralyzed limbs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Drug-induced pemphigoid                                             | <b>Furosemide (#1), ACE inhibitors, cephalosporins, β-lactams, D-penicillamine</b> , Sulfasalazine, NSAIDs, neuroleptics, gold, SSKI, bumetanide, phototherapy<br>Mnemonic: " <b>Fat Abdomens Covered By Pemphigoid = Furosemide, ACE-inhibitors, Cephalosporins, β-lactams, Penicillamine</b> "                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Anti-p200 pemphigoid                                                | Most often p/w classic BP eruption (>DH, eczematous presentations); head and mucous membranes more frequently involved; often <b>a/w psoriasis</b> ; target antigen: <b>laminin γ1</b> ; salt-split skin: IgG binding to <b>dermal</b> side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Anti-p105 pemphigoid                                                | Extensive blistering and denudation on both mucous membranes and skin, resembling <b>SJS/TEN</b> ("p105 = <b>TEN</b> "); target antigen: 105 kDa protein; salt-split skin: IgG binding to <b>dermal</b> side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |





**Fig. 30.6 Bullous pemphigoid – uncommon clinical variants. A,B** In dyshidrosiform pemphigoid, clusters of vesicles and bullae appear on acral skin and can resemble dyshidrotic eczema or pompholyx. **C** In pemphigoid vegetans, vegetating plaques can develop in major body folds, including the inguinal crease. **D** Toxic epidermal necrolysis-like lesions with large erosions.



**Fig. 30.7 Childhood bullous pemphigoid.** **A** Generalized tense bullae and circular crusted erosions. **B** Tense vesicles and bullae in an annular or figurate array at the edge of expanding lesions, a presentation that might be diagnosed clinically as linear IgA bullous dermatosis. **C** A predominance of acral involvement is often seen in infants. *B,C, Courtesy, Julie V Schaffer, MD.*



**Fig. 30.8 Bullous pemphigoid localized to a psoriatic plaque.** No obvious trigger was detected, as the patient was not receiving phototherapy. *Courtesy, Jean L. Bologna, MD.*





- Gestational pemphigoid (pemphigoid gestationis, herpes gestationis, HG):
- 
- This variant occurs during pregnancy (2nd or 3rd trimester usually)
- 
- Peri-umbilical and abdominal region —»• pruritic papular and urticarial lesions ■=> vesicles and bullae (may become generalized)

N.B ???

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| DERMATOSES OF PREGNANCY – FETAL RISK, INVOLVEMENT OF NEWBORN SKIN, AND RISK OF RECURRENCE |                                                                                                             |                                                                                |                                                                                                                     |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Dermatosis                                                                                | Fetal risk                                                                                                  | Newborn skin Involvement                                                       | Risk of recurrence                                                                                                  |
| Pemphigoid gestationis                                                                    | Increased risk of prematurity and small-for-gestational age neonates; risk correlates with disease severity | Mild and transient lesions of pemphigoid gestationis in up to 10% (see Ch. 34) | Commonly recurs ("skipped" pregnancies in only 5–8% of women); recurrences induced by oral contraceptives in 25–50% |
| Polymorphic eruption of pregnancy                                                         | None                                                                                                        | None                                                                           | Usually does not recur                                                                                              |
| Intrahepatic cholestasis of pregnancy                                                     | Increased risk of premature labor (20–60%), intrapartal fetal distress (20–30%) and stillbirths (1–2%)      | None                                                                           | Recurrence in 45–70% of subsequent pregnancies; may be triggered by oral contraceptives                             |
| Atopic eruption of pregnancy                                                              | None                                                                                                        | None                                                                           | Commonly recurs due to atopic diathesis                                                                             |

**Table 27.2** Dermatoses of pregnancy – fetal risk, involvement of newborn skin, and risk of recurrence.



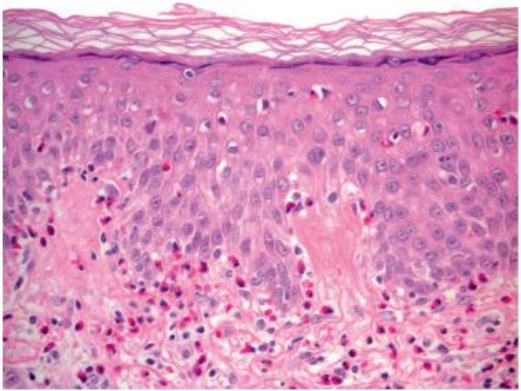
# ASSOCIATION

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1. Internal malignancy :digestive tract, urinary bladder, lung) and lymphoproliferative disorders.
2. Autoimmune disorders: IBD , RA ,LE , Hashimoto's thyroiditis, .
3. psoriasis (anti-p200 pemphigoid) and lichen planus (LPP).

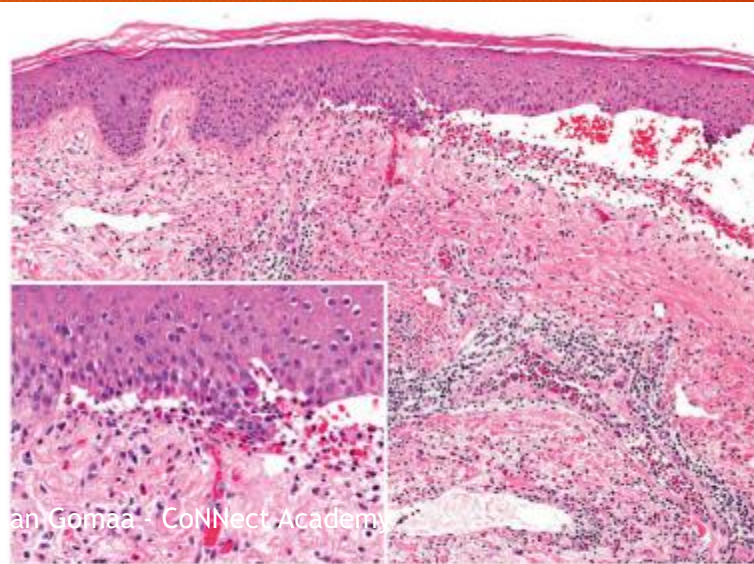
- DRUG: FACA FE GPC فكه في جيبك
  - Furosimide ,
  - Analgesic
  - Captopril
  - Amoxicillin
  - Gold
  - K iodid
  - ciprofloxacin



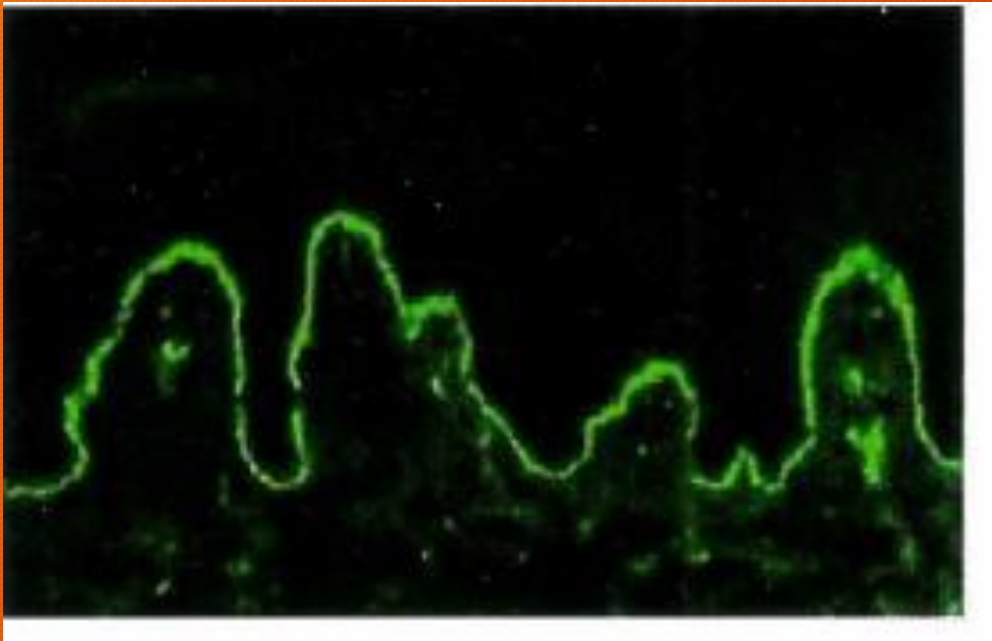


**Fig. 30.9 Urticarial phase of bullous pemphigoid – histologic features.** Eosinophils are present within the dermis as well as the epidermis (eosinophilic spongiosis). Some of the eosinophils have lined up at the dermal–epidermal junction, a typical finding in the urticarial stage of BP. Courtesy, Lorenzo Cerroni, MD.

- Histopathology
  - In the non-bullous phase: eosinophilic spongiosis , dermal infiltrates of eosinophils
  - In the bullous phase (early small intact vesicle): sub-epidermal blister with eosinophils



**Fig. 30.10 Bullous pemphigoid – histologic features.** Subepidermal blister which contains fibrin, eosinophils and mononuclear cells (see inset). Courtesy, Lorenzo Cerroni, MD.



- DIF :linear deposition of igG , C3 at DEJ
- Salt-split skin (SSS) indirect IF: Linear deposits at the BMZ.
  - Autoantibodies bind to the epidermal side (typically)
  - A both the epidermal and dermal sides (less frequently)



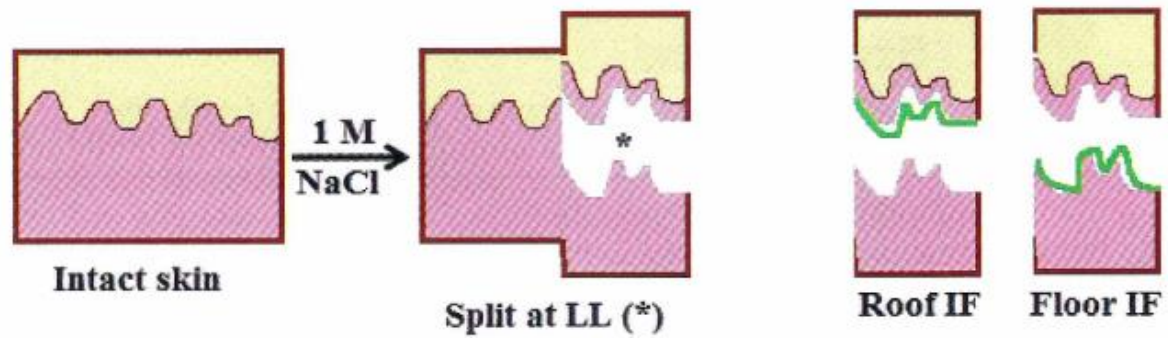
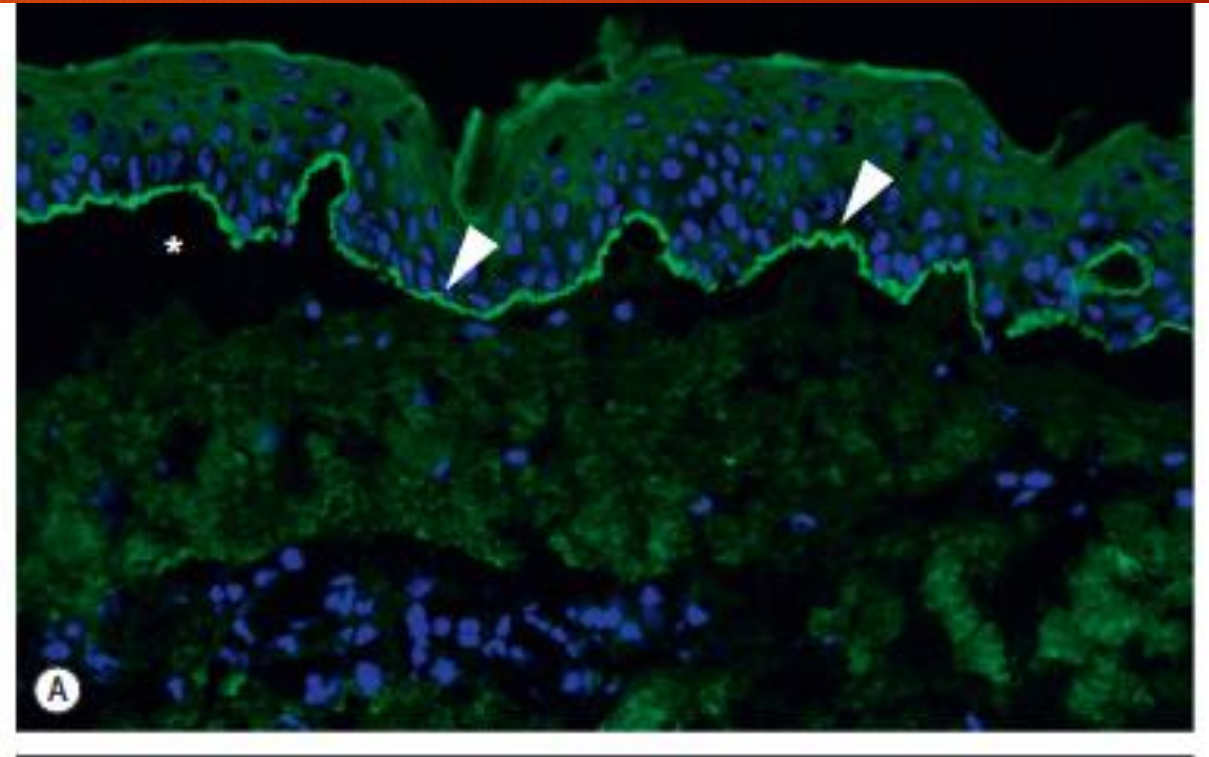


Fig 9.5 Salt-split skin (SSS) immunofluorescence technique



- Non-bullous phase (non-specific)
  - urticaria
  - allergic urticarial reactions
  - urticarial drug reactions
  - urticarial vasculitis
  - contact dermatitis
  - arthropod reactions

#### Bullous phase

- Other sub-epidermal immunobullous disorders
- The pemphigus group
- bullous arthropod bites
- Stevens-Johnson syndrome
- bullous drug eruptions
- pompholyx
- pseudoporphyria or porphyria cutanea tarda
- In children T bullous impetigo, inherited epidermolysis bullosa, bullous mastocytosis



# TREATMENT

Rx

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## THERAPEUTIC LADDER FOR BULLOUS PEMPHIGOID

### Mild and/or localized disease

#### First-line

Superpotent topical corticosteroids (1\*)

#### Second-line

Oral corticosteroids (1)

Minocycline, doxycycline or tetracycline, alone or in combination with nicotinamide (1)

Erythromycin, penicillins (3)

Dapsone, sulfonamides (3)

Topical immunomodulators (e.g. tacrolimus) (3)

### Extensive/persistent cutaneous disease

#### First-line, as primary treatment

Superpotent topical corticosteroids (1\*)

Oral corticosteroids† (1‡)

#### Second-line, or as adjunctive therapy

Azathioprine (2)

Mycophenolate mofetil (2)

Methotrexate§ (2)

Chlorambucil (3)

Cyclophosphamide (3)

IVIg (3)

Plasma exchange (2)

Rituximab (3)

Omalizumab (3)

Immunoadsorption (3)

*Note: Superpotent topical corticosteroids should be considered in any patient and may be combined with a systemic therapy.*

|                                       |   |                                                                               |
|---------------------------------------|---|-------------------------------------------------------------------------------|
| <b>Anti-p105 pemphigoid (105 kDa)</b> | } | Similar to BP but autoantibodies targeting distinct antigens of epidermal BMZ |
| <b>Anti-p200 pemphigoid (200 kDa)</b> |   |                                                                               |



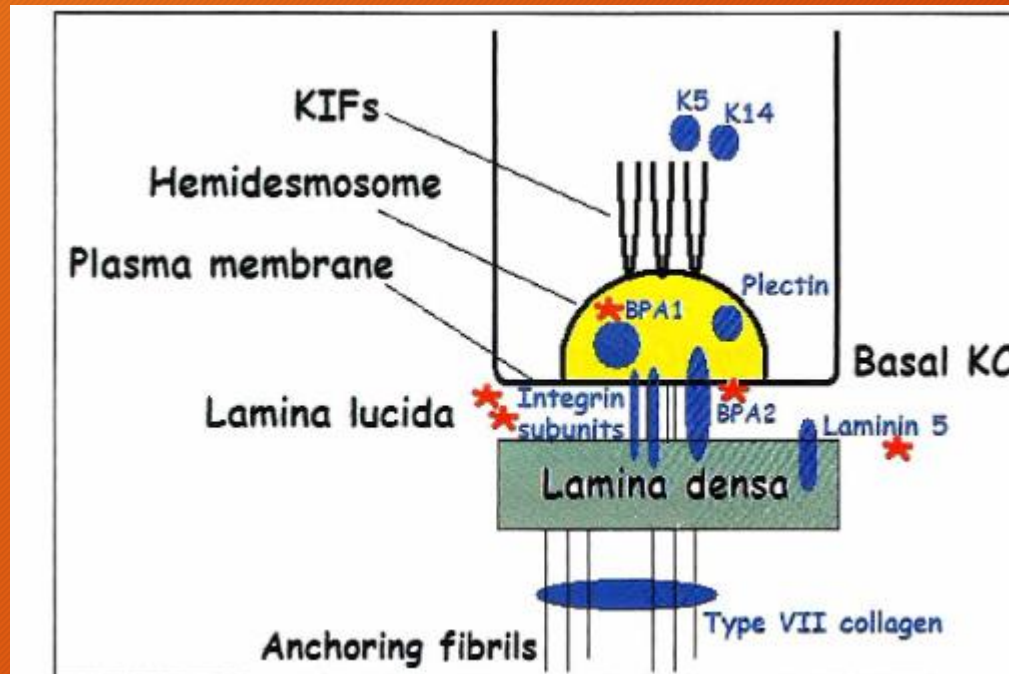
# MUCOUS MEMBRANE (CICATRICAL) PEMPHIGOID

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**Fig. 30.13** Mucous membrane (cicatricial) pemphigoid. Desquamative gingivitis with erythema and erosions of the gingival margins. Note the sloughing of the mucosa with shaggy margins.

- a rare autoimmune sub-epithelial blistering disorder characterized by
- 1. predominant involvement of the mucosae (may affect the skin),
  2. tendency towards scarring
  3. chronic and progressive course
- complications e.g. atrophic scarring and fibrosis of the conjunctivae and blindness).
- Elderly women



- Autoantibody against
- Laminine 5
- Integrine
  - alpha 6 : oral
  - , beta 4 : ocular
- BPA1 :mucosa , skin
- BPA2 :distal part :mucosa , skin





**Fig. 30.14** Mucous membrane (cicatricial) pemphigoid. Chronic erosions on the hard palate with irregular borders. Sloughing of mucosa is seen superiorly.

- 
- Mucoa
- Oral 85 %: may be the only site
  - Desquamative gingivitis , erosions
  - Affection palate , buccal mucosa
  - Healing with white stricks , scar
  - periodontal ligam ent damage and loss of teeth.



**Fig. 30.15 Mucous membrane (cicatricial) pemphigoid progression.** A Erosion and erythema of the lower medial eyelid margin plus scale-crust of the inner canthus and lower eyelid. B Three months later, ectropion and thickening of the lower eyelid in addition to erosions. C Six months later, smaller erosions but scarring and milia formation. D Seven years later, further scarring with significant shortening of the inferior fornix due to symblepharon. Courtesy, Louis A Fragola, Jr, MD.



**Fig. 30.16 Mucous membrane (cicatricial) pemphigoid.** Typical ocular involvement as manifested by fibrous tracts, representing partial or incomplete symblepharon.

- **Conjunctiva : common**
  - Nonspecific chronic conjunctivitis , burning , soreness>>>
  - Conjunctival Fibrosis ,blindness >>>> blindness
- **Others :pharynx , larynx , oesophagus**
  - Skin : 25-30% scalp , face , neck , upper trunk
  - Erythematous plaque >>> blistering >> rupture >>>scarring



- Brunsting -perry pemphigoid
  1. Blisters >>head , neck >>>
  2. Cicatricial alopecia
  3. No mucosa



**Fig. 30.17 Mucous membrane (cicatricial) pemphigoid – Brunsting–Perry variant.** **A** Crusted erosion on the lower cheek within an oval area of inflammation and scar. **B** Involvement of the scalp with scarring and atrophy resulting in alopecia. *B, Courtesy, Karynne O Duncan, MD.*

- H.P:
  - Sub-epidermal or sub-epithelial blister without acantholysis.  
A mixed infiltrate
  - older lesions A fibrosis in the upper dermis.
  - EM>>> cleavage occurs within the LL.
- DIF : (peri-lesional, uninvolved skin)
  - IgG and/or C3 along the epithelial BMZ (fine, linear, continuous).
- 
- DIF of mucosae are more frequently positive than those of skin.
- 
- (SSS) in DIF : Autoantibodies bind to a
  - A the epidermal roof (most patients).
  - A the dermal side (anti laminin 5).



# Treatment

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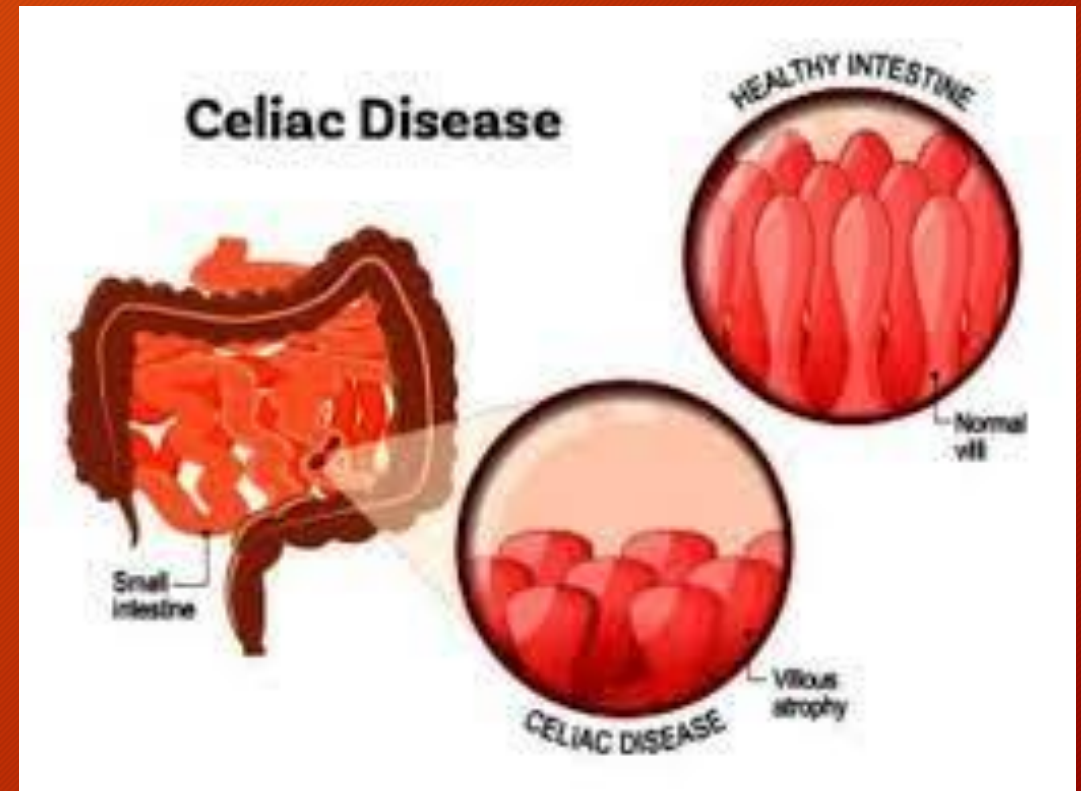
- Mild : topical superpotent steroid
- Sever oral ,cutaneus :**DAPSON** 1<sup>st</sup> line treatment (50 -150 mg/kg)
- Sever ocular : cuclophosphamide
- Oesophageal , laryngeotrachial : cyclophosphamide +steroid +mff
- Surgical :sever scaring

# Dermatitis herpetiform (duhring disease )

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**Fig. 31.3 Dermatitis herpetiformis.**  
**A** Grouped pink papulovesicles on the upper back, neck and scalp in a child. **B** Pruritic pink papules on the buttocks, some of which have central hemorrhagic crusts. *ij*  
*Courtesy, Louis A. Frangola, Jr, MD.*





- DH is a cutaneous manifestation of gluten-sensitive enteropathy (GSE) / celiac disease
- DH associated GSE in all cases but symptomatic in 20% of cases only

# Aetiopathogenesis

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Enviromental  
triggers

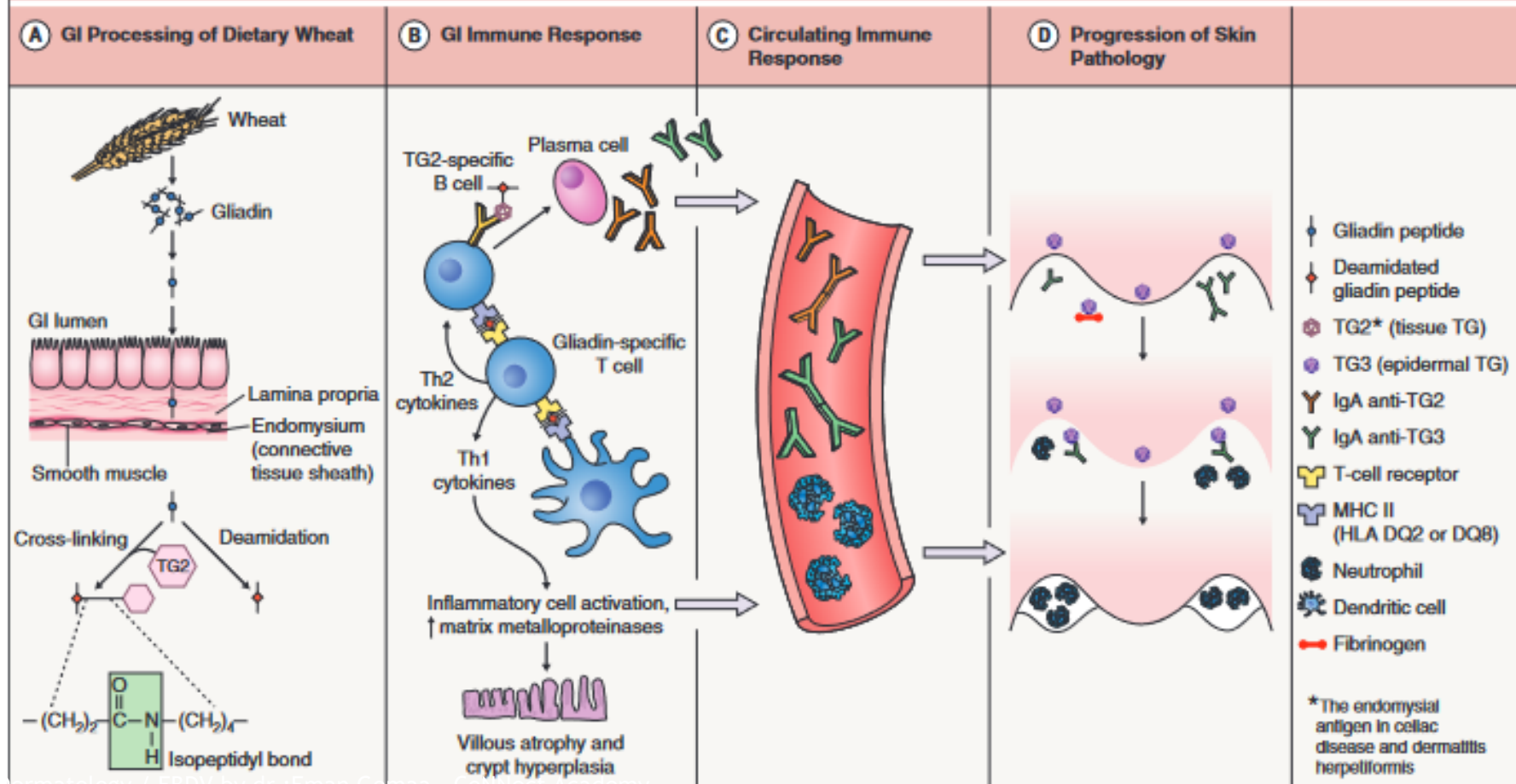
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Genaticly  
susptable pt

Immun  
e  
dysregulation



- Genetic :
  - HLA DQ2 , DQ8
- Environmental :
  - Gluten
  - Iodide

# PROPOSED PATHOGENESIS OF DERMATITIS HERPETIFORMIS AND CELIAC DISEASE





# Clinical feature

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- Symptom : Intense pruritis
- Sign : polymorphic
  - Papulovesicle on erythematous base >>> excoriation
  - Bilateral , symmetrical ,
  - Extensor , buttocks , scalp

### AUTOIMMUNE DISORDERS ASSOCIATED WITH DERMATITIS HERPETIFORMIS

#### Common

- Autoimmune thyroid disease (Hashimoto thyroiditis)
- Insulin-dependent diabetes mellitus

#### Uncommon

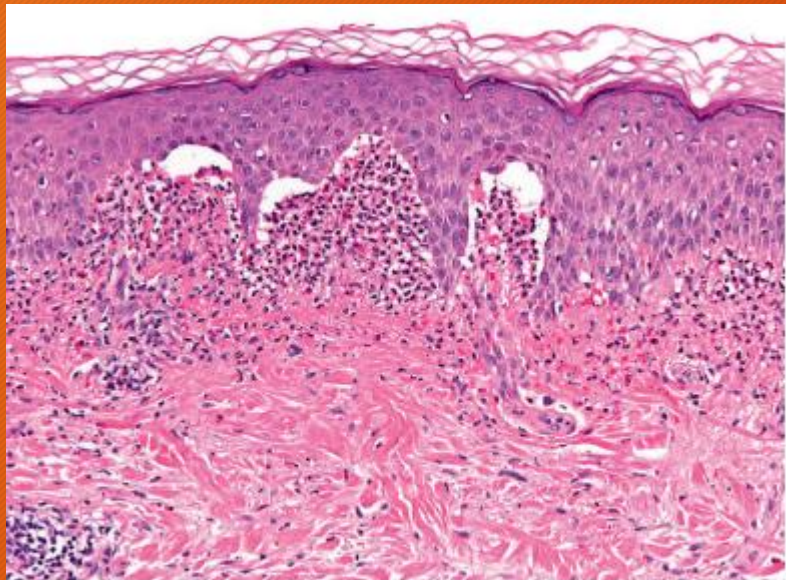
- Pernicious anemia

#### Rare

- |                                       |                                    |
|---------------------------------------|------------------------------------|
| • Addison disease                     | • Systemic sclerosis (scleroderma) |
| • Autoimmune chronic active hepatitis | • Sjögren syndrome                 |
| • Alopecia areata                     | • Systemic lupus erythematosus     |
| • Myasthenia gravis                   | • Vitiligo                         |
| • Sarcoidosis                         |                                    |

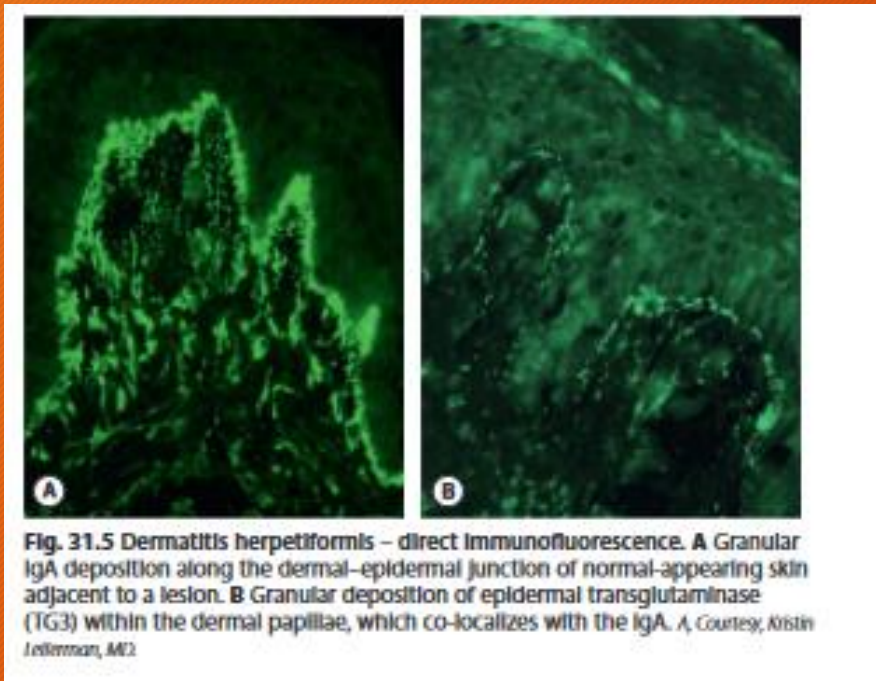
- Association :
  - Autoimmune hypothyroidism
  - Malignancy : entropathy associated t cell lymphoma





**Fig. 31.4 Dermatitis herpetiformis – histopathologic features.** Subepidermal clefts beneath which are collections of neutrophils within dermal papillae. Scattered eosinophils are also present. *Courtesy, Lorenzo Cerroni, MD.*

- sub-epidermal clefts with collections of neutrophils within dermal papillae (papillary microabscesses)



- DIF :Findings: Granular IgA deposits localized to dermal papillae.
- IIF :NEGATIVE
  - IgA/TG3 immune complexes are formed locally within the papillary dermis.



# TREATMENT

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- Gluten-free diet includes corn, rice and oats. Patients should AVOID wheat, barley and rye
- DAPSON :DRAMATIC RESPONSE
  - Initial dose :25-50 mg daily
- Topical steroid
- antihistaminic

# Linear igA bulloue dermatosis

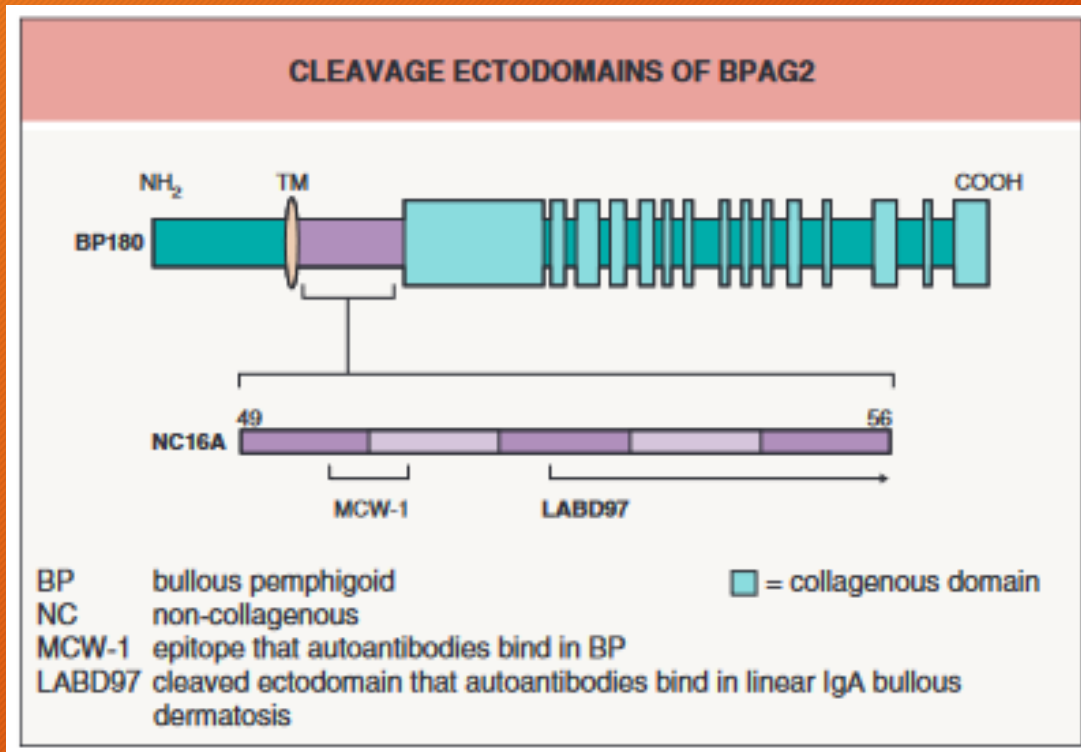
44

LABD is an autoimmuno sub-epidermal blistering disease.

Adult 6 decade : Labd

Child 4-5 years :cbdc





- 2 types :
  - Lamina lucida type :BPA2 (LABD97)
  - Sub lamina densa: type 7 collagen

# Clinical features

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**Fig. 31.12 Linear IgA bullous dermatosis.** Annular and herpetiform vesicles arising on an inflammatory base. Annular pink plaques are also present. *Courtesy, Jeffrey P Callen, MD.*

- Adult :as DH , BP
  - Herpiform arrangement , or annular plaque
  - Flexors
  - Koebner phenomena





**Fig. 31.7 Linear IgA bullous dermatosis.** Characteristic findings in this child include the annular array of bullae (inset) and involvement of the genital region. There are also tense bullae arising on normal-appearing skin with either clear or hemorrhagic fluid and annular bullae with central crusting. Courtesy, Antonio Iorello, MD.

- Children :
  - Annular erythematous blister >>> crown of jewels
  - Flexural area , lower trunk , thigh
- LABD may present as a variant of mucous membrane (cicatricial) pemphigoid,
- with oral, nasal, pharyngeal and esophageal lesions.

- Association :
  - Gluten sensitive enteropathy
  - autoimmune , malignancy
- Drugs association :
  - **VANCOMYCIN**
  - Peniciline
  - NSAID
  - Cyclosporin



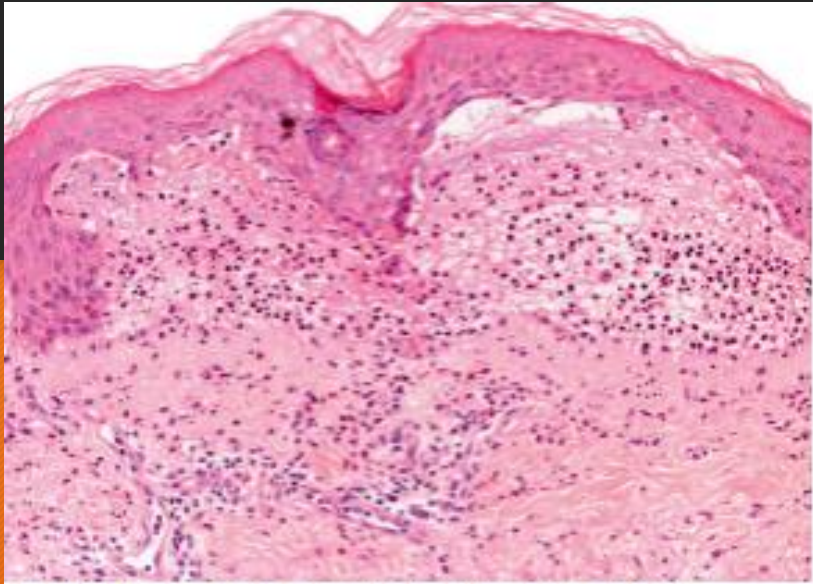


Fig. 31.14 Linear IgA bullous dermatosis – histopathologic features. Subepidermal blister filled with neutrophils. Neutrophils and a few eosinophils are also present within the underlying dermis. Courtesy, Lorenzo Giamari, MD.

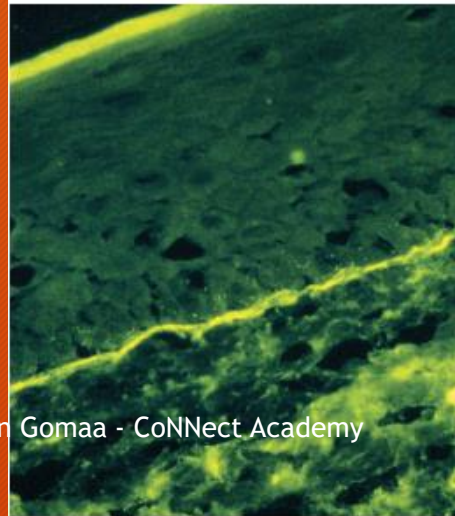


Fig. 31.8 Linear IgA bullous dermatosis – direct immunofluorescence. A linear pattern of IgA deposition is present within perilesional

- H.P :
  - Sub-epidermal blister which neutrophils in the underlying dermis (with /without eosinophils.
- Direct IF: (peri-lesional, uninvolved skin)
  - Linear IgA deposition along the BMZ
- Indirect IF:
  - Circulating antibodies of the IgA class can be demonstrated in 60-70% of LABD sera
- SSS:
  - Lamina lucida type o f LABD —> adhere to the epidermal side (roof).
  - Sub-lamina densa type of LABD —> adhere to the dermal side (floor).

# treatment

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1. DAPSON :100 :300 mg daily
2. Oral prednisone :40 mg



| SIDE EFFECTS OF DAPSONE           |                                                                                                                                                                                             |                                                                                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Red blood cell toxicity           | <ul style="list-style-type: none"> <li>• Hemolytic anemia</li> <li>• Methemoglobinemia<sup>^</sup></li> </ul>                                                                               |                                                                                                                                            |
| White blood cell toxicity         | <ul style="list-style-type: none"> <li>• Leukopenia</li> <li>• Agranulocytosis</li> </ul>                                                                                                   |                                                                                                                                            |
| Dapsone hypersensitivity syndrome | <ul style="list-style-type: none"> <li>• Fever, fatigue, anorexia, hepatitis, lymphadenopathy</li> </ul>                                                                                    |                                                                                                                                            |
| Cutaneous reactions               | <ul style="list-style-type: none"> <li>• Morbilliform eruption</li> <li>• Urticaria</li> <li>• Fixed drug eruption</li> <li>• Erythema nodosum</li> <li>• Exfoliative dermatitis</li> </ul> | <ul style="list-style-type: none"> <li>• SJS/TEN</li> <li>• Phototoxicity</li> <li>• Drug-induced cutaneous lupus erythematosus</li> </ul> |
| Gastrointestinal manifestations*  | <ul style="list-style-type: none"> <li>• Anorexia, nausea</li> <li>• Hepatitis</li> <li>• Cholestatic jaundice</li> <li>• Severe hypoalbuminemia</li> </ul>                                 |                                                                                                                                            |
| Neurologic associations*          | <ul style="list-style-type: none"> <li>• Headache, dizziness</li> <li>• Peripheral neuropathy</li> <li>• Blurred vision, tinnitus</li> <li>• Insomnia</li> <li>• Psychosis</li> </ul>       |                                                                                                                                            |
| Miscellaneous                     | <ul style="list-style-type: none"> <li>• Fever</li> <li>• Nephrotic syndrome</li> </ul>                                                                                                     |                                                                                                                                            |

| CHARACTERISTICS THAT DIFFERENTIATE DH, LABD, AND BP |                                                      |                                                                                      |                                                   |
|-----------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------|
|                                                     | DH                                                   | LABD                                                                                 | BP                                                |
| Cutaneous lesion                                    | Grouped papules and small vesicles, often excoriated | Small vesicles and/or large bullae                                                   | Large tense bullae                                |
| Distribution pattern                                | Extensor surfaces, symmetrical                       | Similar to DH or BP                                                                  | Trunk, extremities, occasionally mucosal surfaces |
| Routine histology                                   | Subepidermal bullae with neutrophilic infiltrate     | Subepidermal bullae with neutrophilic infiltrate (sometimes eosinophils predominate) | Subepidermal bullae with eosinophilic infiltrate  |
| Direct IF                                           | Granular IgA in dermal papillae                      | Linear IgA at BMZ, possibly also IgG                                                 | Linear IgG and C3 at BMZ                          |
| Site to biopsy for direct IF                        | Adjacent normal-appearing skin                       | Perilesional                                                                         | Perilesional                                      |
| Routine indirect IF                                 | Negative                                             | Linear IgA at BMZ (70%)                                                              | Linear IgG at BMZ (70%)                           |
| Enteropathy                                         | >90%                                                 | Rare                                                                                 | None                                              |
| HLA-DQ2                                             | >90%                                                 | 30%                                                                                  | Normal (20%)                                      |
| Dapsone responsiveness                              | Excellent                                            | Good, may also require systemic corticosteroids                                      | Minimal to moderate                               |



# Epidermolysis bullosa acquista

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- EBA is a rare acquired, sub-epidermal immuno-bullous disease
- type V II collagen (anchoring fibrils) in the SLD.
- EBA has 2 main clinical presentations:
  1. Mechano-bullous “Non-inflammatory” disorder (resembling dystrophic EB) ■=> the classic presentation
  2. Immuno-bullous “Inflammatory” disorder (indistinguishable from BP, CP)



**Fig. 30.18 Epidermolysis bullosa acquisita – mechanobullous presentation.** Milia and scarring that favor sites of trauma overlying joints, in association with skin fragility. Note the resemblance to dystrophic epidermolysis bullosa.

- Type 1 mechanobullous :
  - Acral blister >>erosion >>>atrophic scar >>>milia , mutilation
  - Mitten deformaty





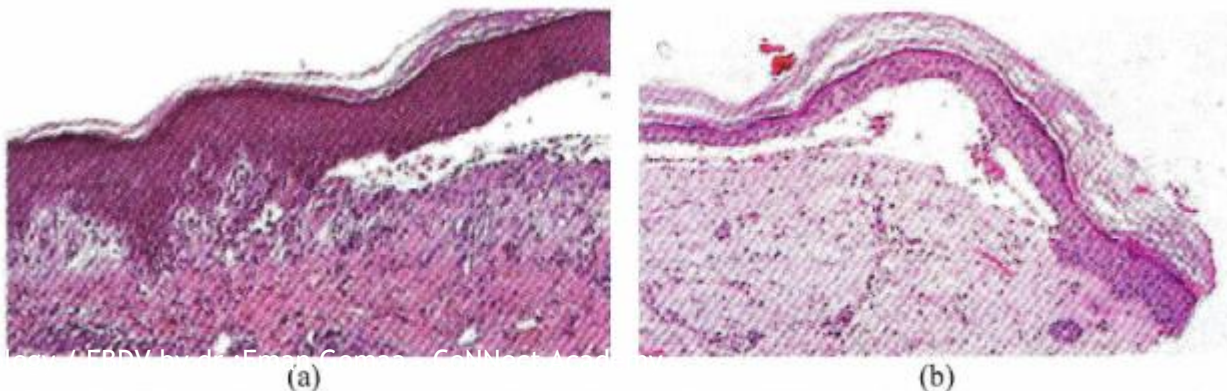
Fig. 30.19 Epidermolysis bullosa acquisita – inflammatory bullous pemphigoid-like presentation. Bullae and circular erosions in a patient with multiple myeloma.



Fig. 30.20 Epidermolysis bullosa acquisita – oral involvement. Multiple erosions of the palate reminiscent of mucous membrane (cicatrical) pemphigoid. Courtesy: C. Probst, MD.

- Type 2 :immune bullouse
  - BP like :widespread vesicles and bullae, involving intertriginous and flexural areas, heal without milia or atrophic scars
  - Cp like (brunsting perry)
  - mucosa :variable
- Association :crohns disease

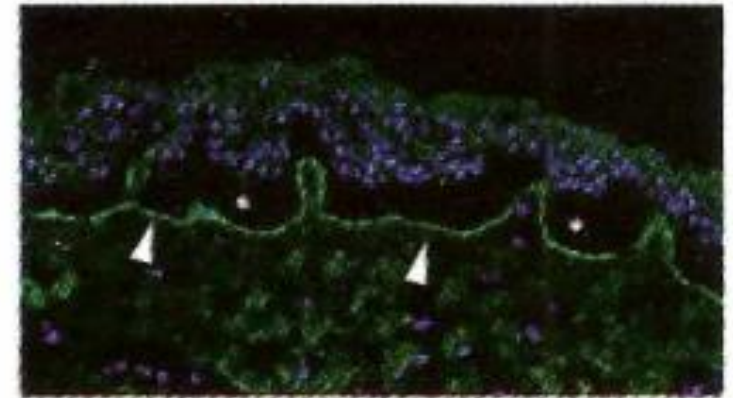
- H.p :subepidermal cleavage
  - BP-like or CP-like variants ■=> mixed inflammatory infiltrate of neutrophils, eosinophils or lymphocytes
  - mechanobullous, non-inflammatory lesions ■=> minimal-to-absent cellular infiltrate



**Fig 12.2 Histopathological features of EBA** Sub-epidermal blister (a) Inflammatory type: mixed inflammatory infiltrate. (b) Non-inflammatory type: minimal inflammatory infiltrate.



- Direct IF :
  - IgG along the epithelial BMZ(broad, linear, continuous)
- Salt-split skin (SSS) indirect IF:
  - Autoantibodies bind to the dermal side(floor)



**Fig 12.3 IIF utilizing SSS in EBA.** Auto-antibodies react with the dermal side (floor) of the blister (arrows).

- The treatment of EBA is difficult and often unsatisfactory
  1. Systemic corticosteroids
  2. Standard immunosuppressive agents e.g. azathioprine, methotrexate,



# Bullous SLE ??????

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**Table 3-7.** Subepidermal Blistering Disease Characteristics

| Disease                                      | Antigen                                                                                              | Size (kDa)        | DIF                                                                                               | Salt-Split Skin                                                       |
|----------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Bullous pemphigoid                           | BPAG1 (plakin)<br><b>BPAG2 (collagen XVII)</b>                                                       | 230<br><b>180</b> | <b>Linear C3 and IgG</b> along BMZ in<br>"n-serrated" pattern                                     | Epidermal                                                             |
| Pemphigoid gestationis                       | BPAG2 (Collagen XVII)                                                                                | 180               | <b>Linear C3 &gt; IgG</b> along BMZ                                                               | Epidermal                                                             |
| LABD                                         | <b>LAD-1 (120kD</b> cleaved<br>portion of BPAG2)<br><b>LABD97 (97kD</b> cleaved<br>portion of LAD-1) | 120→ <b>97</b>    | Linear IgA +/-C3 along BMZ                                                                        | Epidermal (IgA)                                                       |
| Mucous membrane<br>pemphigoid (classic form) | BPAG2 ( <b>C-terminus</b> )                                                                          | 180               | Linear IgG and C3 along BMZ                                                                       | Epidermal (or both sides with<br>stronger staining on epidermal side) |
| Ocular-predominant MMP                       | <b>β<sub>4</sub> integrin</b>                                                                        | NA                | Linear IgG and C3 along BMZ                                                                       | Epidermal                                                             |
| Anti-epiligrin MMP                           | <b>Laminin 332</b>                                                                                   | 400-440           | Linear IgG and C3 along BMZ                                                                       | Dermal                                                                |
| p200 pemphigoid                              | Laminin γ1                                                                                           | 200               | Linear IgG and C3 along BMZ                                                                       | Dermal                                                                |
| p105 pemphigoid                              | NA                                                                                                   | 105               | Linear IgG and C3 along BMZ                                                                       | Dermal                                                                |
| EBA                                          | <b>Type VII collagen</b><br>(anchoring fibrils)                                                      | 290               | <b>Linear IgG &gt; C3</b> along BMZ in "u-serrated"<br>pattern                                    | Dermal                                                                |
| Bullous SLE                                  | <b>Type VII collagen</b><br>(anchoring fibrils)                                                      | 290               | <b>Granular to linear</b> staining w/ <b>multiple<br/>reactants</b> (IgG, IgA, IgM, C3)           | Dermal                                                                |
| PCT                                          | NA                                                                                                   | NA                | <b>Linear IgG (&gt;IgM), C3 and fibrinogen</b> along<br>BMZ and <b>around superficial vessels</b> | Negative                                                              |



**Table 3-9.** Other Blistering Disorders

| Disease                              | Clinical Setting                                                                                                           | Clinical Presentation                                                                                                                                                                                                                                                                                                                                                                                                                    | Blister Location                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Bullous diabeticorum                 | Longstanding diabetics w/ peripheral neuropathy, retinopathy, or nephropathy                                               | p/w sudden eruption of <b>tense, non-inflammatory</b> , clear fluid-filled vesicles/ bullae, 0.5 to several cm, on <b>feet</b> (>lower legs > hands > forearms); histopathology: <b>cell-poor subepidermal blister</b> ; DIF negative; Rx: spontaneous resolution in 2–6 wks; may aspirate if uncomfortable                                                                                                                              | Subepidermal                                                                    |
| Coma blister                         | Coma due to meds ( <b>barbiturates</b> > <b>benzos</b> , alcohol, opioids), or non-drug-induced coma                       | <b>Tense blisters</b> arising on <b>sites of pressure</b> 48–72 hrs after loss of consciousness; blisters due to pressure-induced necrosis; histopathology: cell-poor subepidermal blister w/ <b>sweat gland necrosis</b> +/-epidermal necrosis; DIF negative; Rx: heals spontaneously in 1–2 wks                                                                                                                                        | Subepidermal                                                                    |
| Friction blisters                    | New pair of <b>ill-fitting shoes</b> ; sports or military involvement                                                      | Very common in <b>young, physically active</b> , repetitive friction; initially red macules at site of friction → painful <b>intraepidermal blisters</b> w/blood-tinged fluid; histopathology: pauci-inflammatory <b>blister just under granular layer</b> ; DIF negative; Rx: heals spontaneously; may aspirate to relieve pressure                                                                                                     | Intraepidermal                                                                  |
| Bullous small vessel vasculitis      | Pts w/small vessel vasculitis (cutaneous or systemic)                                                                      | LCV w/superimposed hemorrhagic vesicles and bullae, usually on <b>distal extremities</b> → may ulcerate; histopathology: <b>LCV w/massive subepidermal edema/bullae</b> and epidermal necrosis                                                                                                                                                                                                                                           | Subepidermal                                                                    |
| Bullous drug eruptions               | Pts receiving meds                                                                                                         | See <a href="#">Table 3-10</a>                                                                                                                                                                                                                                                                                                                                                                                                           | See <a href="#">Table 3-10</a>                                                  |
| Bullous Arthropod Assault            | Most common in children or pts with <b>hematologic malignancy</b> ( <b>CLL</b> > mantle cell lymphoma, NK/T-cell lymphoma) | <b>Grouped pruritic papules</b> w/central blistering; may have persistent papulonodules (persistent arthropod assault); <b>pts w/lymphoma may have reaction in absence of definitive insect bite (“bug bite-like reactions”)</b> ; histopathology: eosinophilic spongiosis, <b>superficial and deep PV/PA lympho-eosinophilic inflammation</b> (+/- flame figures ), superficial dermal edema; Rx: topical steroids, oral antihistamines | <b>Intraepidermal</b> > subepidermal blister (may occur if dermal edema severe) |
| Delayed postburn/ postgraft blisters | Blisters at sites of prior trauma (burns, graft site)                                                                      | Tense vesicles/bullae appearing weeks to months (avg 37 days) after original wound has completely healed; <b>due to fragility of new DEJ</b> ; histopathology: cell-poor subepidermal blister; negative DIF; Rx: spontaneous resolution                                                                                                                                                                                                  | Subepidermal                                                                    |
| Edema blisters                       | Anasarca or acute exacerbation of chronic edema                                                                            | <b>Tense</b> blisters in edematous dependent sites ( <b>distal LE, feet</b> most common); histopathology: cell-poor subepidermal bullae with massive dermal edema and epidermal spongiosis; Rx: treat underlying edema, compression wraps                                                                                                                                                                                                | Subepidermal                                                                    |



**Table 3-10.** Bullous Drug Eruptions

| Disease                                                                                                                                                                                                                                                                                                                                                                                  | Characteristic Features                                                                                                                                                                                                                        | Commonly Implicated Drugs                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed drug eruption                                                                                                                                                                                                                                                                                                                                                                      | Sharply circumscribed <b>erythematous to dusky</b> violaceous patches<br>Central blisters or erosions may appear<br>Often resolves with <b>postinflammatory hyperpigmentation</b><br>Recurrence at same location(s) following drug re-exposure | <b>Sulfonamides, NSAIDs, tetracyclines</b> , barbiturates, aspirin, acetaminophen (paracetamol), metronidazole, phenolphthalein                                                                                                                                     |
| Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN)                                                                                                                                                                                                                                                                                                                         | Prodrome of fever and <b>painful skin</b><br>Areas of dusky erythema associated with epidermal detachment (varying from <10% [SJS] to >30% [TEN])<br>Mucosal involvement                                                                       | NSAIDs, antibiotics (sulfonamides and $\beta$ -lactams), anticonvulsants, allopurinol                                                                                                                                                                               |
| Drug-induced autoimmune blistering diseases                                                                                                                                                                                                                                                                                                                                              | Primarily linear IgA bullous dermatosis, pemphigus, bullous pemphigoid<br>Diagnosis based on histologic findings, immunofluorescence studies, and drug history                                                                                 | <u>Linear IgA bullous dermatosis</u> : <b>vancomycin</b> > $\beta$ -lactams, captopril, NSAIDs<br><u>Pemphigus</u> : <b>penicillamine, captopril</b> , $\beta$ -lactams, gold<br><u>Bullous pemphigoid</u> : diuretics (especially <b>furosemide</b> ), antibiotics |
| Drug-induced pseudoporphyria                                                                                                                                                                                                                                                                                                                                                             | Eruption resembles porphyria cutanea tarda<br>Porphyrin determinations are within normal limits                                                                                                                                                | NSAIDs (especially <b>naproxen</b> ), nalidixic acid, thiazides, furosemide, tetracyclines                                                                                                                                                                          |
| Acute generalized exanthematous pustulosis                                                                                                                                                                                                                                                                                                                                               | Acute onset, usually occurring within 2 days of drug exposure<br>Areas of erythema studded with pustules; occasionally vesicles<br>Fever, malaise, leukocytosis                                                                                | <b><math>\beta</math>-lactams</b> , macrolides, pristinamycin, terbinafine, calcium channel blockers (diltiazem), hydroxychloroquine, carbamazepine, acetaminophen, metronidazole                                                                                   |
| Phototoxic drug eruptions                                                                                                                                                                                                                                                                                                                                                                | Limited to sun-exposed areas<br>Resembles exaggerated sunburn                                                                                                                                                                                  | <b>Tetracyclines</b> (especially doxycycline), <b>quinolones</b> , <b>psoralens</b> , NSAIDs, diuretics                                                                                                                                                             |
| Bromoderma and iododerma                                                                                                                                                                                                                                                                                                                                                                 | <b>Acneiform lesions</b> , papulopustules, nodules, or even <b>vegetating lesions</b> simulating pemphigus vegetans<br>Clear or hemorrhagic blisters can develop (more common in iododerma)                                                    | Bromides, iodine-containing drugs (e.g. amiodarone), radiographic contrast media                                                                                                                                                                                    |
| Palmoplantar erythrodysesthesia (acral variant of toxic erythema of chemotherapy)                                                                                                                                                                                                                                                                                                        | Painful erythema develops primarily on the palms, soles and digits following chemotherapy administration<br>The skin becomes edematous, its color changes to dark red or violet, and blisters and erosions may develop                         | <b>Cytarabine</b> , doxorubicin, capecitabine, 5-fluorouracil (especially prolonged infusions), multi-kinase inhibitors (e.g., sorafenib, sunitinib), busulfan, taxanes, clofarabine, pralatrexate                                                                  |
| (From Bologna JL, Jorizzo JL, Rapini RP. Dermatology, 3rd Ed. Elsevier. 2012)<br>Occasionally, eczematous drug reactions (e.g., secondary to warfarin, calcium channel blockers) and systemic contact dermatitis can be papulovesicular; patients with drug reaction eosinophilia and systemic symptoms (DRESS)/drug-induced hypersensitivity syndrome (DIHS) can also develop vesicles. |                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                     |





# Thank you

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